

AP Series

Advanced Performance UniBloc Balances

Provides High-speed Response and High Stability

A New Stage in Analytical Balance Performance

High Speed

The response time for trace measurements (from 1 mg) is about 2 seconds.
This significantly improves weighing efficiency.

Stress Free

The STABLO™-AP ionizer can be mounted.
This eliminates the influence of static electricity, achieving reliable measurements in a simpler procedure.

For Regulation

Interlocking with LabSolutions Balance enables compliance with a variety of data integrity regulations, including ISO 17025 for testing laboratories, ISO 9001 and ISO 14001 for the manufacturing industry, and GLP/GMP and the United States Pharmacopeia (USP) for the pharmaceutical industry.

For HPLC

Functions are included for the preparation of buffer solutions used in HPLC.
As a result, the operation can be performed accurately and easily, even by non-specialists.

Save Your Operation

Equipped with USB as standard*1. Includes many diverse functions to support users.

*1 All models: USB-B type connector as standard AP-W Series: USB-A type and B type as standard



* AP224W with optional ionizer



Watch the AP overview video on our website.
<https://www.shimadzu.com/an/balance/analytical/ap.html>

High Speed

Fast measurement significantly improves operational efficiency.

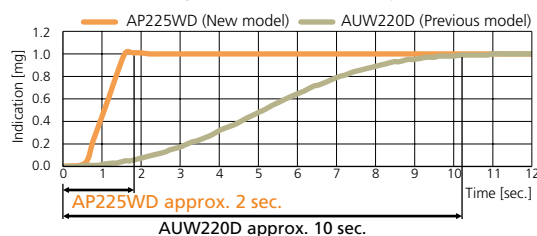
Fast Response with UniBloc AP Technology

Shimadzu analytical balances boast the one-piece UniBloc weighing sensor, which is now even more advanced.

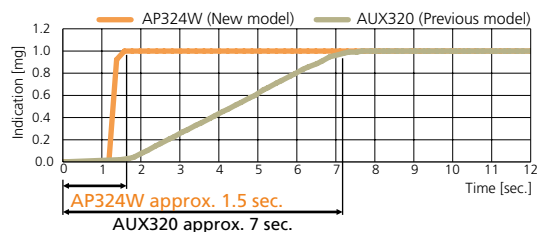
The response time is reduced to about 1/5 the time of previous models.

In addition, the UniBloc sensor offers a response time of just 2 seconds, an improvement from 10 seconds with the previous model.

Response During Trace Measurements with the 0.01 mg Model (Equivalent to 1 mg / With Conditions Set by Shimadzu)



Response During Trace Measurements with the 0.1 mg Model (Equivalent to 1 mg / With Conditions Set by Shimadzu)



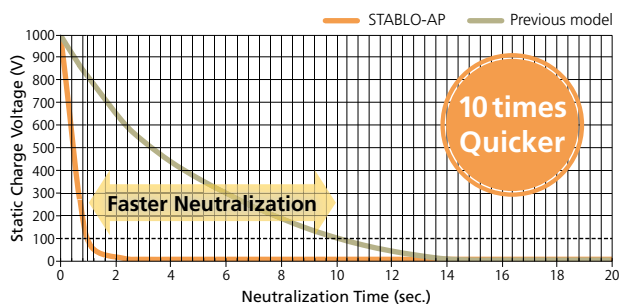
Model	Previous Model	AP Series
0.01 mg	10 sec.	2 sec.
0.1 mg	7 sec.	1.5 sec.



* AP135W

Built-in High-Performance Ionizer (Optional)

Comparison of Neutralization Speed (Representative Values)



Measurement Conditions

- Time from ± 1000 V to ± 100 V
- For this evaluation, a 150×150 mm charged plate monitor (CPM, 20pF) was used.
- Distance between CPM and ionizer: 100 mm

The ionizer eliminates influence of static electricity in 1/10 the time of previous models.

Note: Example of typical static electricity removal time (± 1000 V $\rightarrow$ ± 100 V)
1 sec. for STABLO-AP and 10 sec. for STABLO-EX (previous model)

AC Method with
Excellent Ion Polarity Balance

Mount the STABLO-AP
in the balance and use it
as a built-in model



Stress Free

A variety of accessories and options suitable for semi-micro measurements

Highly Sophisticated Simulation Technology

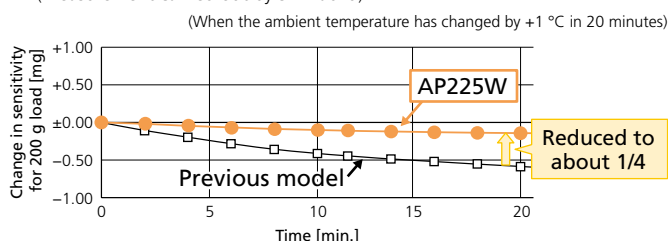
Increased weighing capacity from 135 mg to 220 g (0.01 mg model)

Micro amount weighing over 135 mg (read 0.01 mg step) is possible.

Improved Sensitivity Stability When Ambient Temperature Changes

The temperature of the operational environment is affected by the external air temperature, turning off the air-conditioning, people entering the room, etc. The stability with respect to these small temperature variations in the operational environment has been improved. When the ambient temperature has changed by +1 °C in 20 minutes, the AP225W provides an improvement in the stability of the sensitivity by a factor of four compared to the previous model.

■ Change in Sensitivity When Repeatedly Weighing 200 g (Measurement carried out by Shimadzu)



Improved Minimum Weight

(Minimum display 0.01mg, AP225W/135W/225WD/125WD only)

By improving stability technologies, the minimum weight required for meeting USP Chapter 41 requirements has been improved from 30 mg to 20 mg.

*In a factory test in our company

The AP Holder (standard accessory for AP225W) compatible with a variety of weighing containers is provided standard, and static electricity is properly removed from the bottom of the container, resulting in easier operation.

→ The AP Holder in combination with the ionizer can eliminate the influence of static electricity on the weight value.
See page 23 for more information.



Volumetric flask
(100 mL)



Test tube
(10 mL)



AP Holder

Containers that can be used with the AP Holder (Examples)

Container	Applicable Volume*
Volumetric flask	10 to 100 mL
Conical flask	100 mL
Beaker	
Centrifuge tube (Spitz tube)	3 to 25 mL
Test tube	

* About 70 mm or more height or length is required.

Easy-to-Use Multi Stand

(0.01 mg model only, equipped as standard)



With weighing paper, for example, if the tare is larger than the pan diameter, measurements can be simplified by attaching the special multi stand.



The internal windbreak plate suppresses the influence of convection and air flow within the weighing chamber, improving the stability and response of measurement values.

For HPLC

Buffer Solution Preparation Mode **NEW**

(AP-W Series only)

- **Recipes for 13 commonly used buffer solutions are included**

Preparation recipes for commonly used buffer solutions are provided as standard.

e.g. disodium phosphate, sodium acid citrate

- **New buffer solution recipes can be registered**

If a buffer solution is not registered by default, it can be added.

- **Instructions are shown on the display**

The target weighing value is shown on the display and analog bar in order to compare the target with the current weight. Manual calculation is not needed.

- **Record function**

Record output with date, time and operator name.

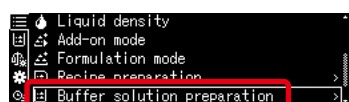
The pH level of mobile phase (eluent) solutions used in liquid chromatographs is adjusted to improve separation of components and extend the life of columns. This pH adjustment process is performed using a buffer solution. Currently, the most common method is using a pH meter to measure the pH as the solution is prepared; however, this process requires considerable time and effort, which can cause operational bottlenecks.

An alternative method does not require a pH meter. It involves preparing solutions by weighing fixed theoretically calculated quantities of an acid and base.

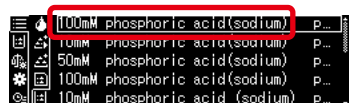
AP series supports weighing these acids and bases. If the type and quantity of the buffer solution are specified, the balance displays the type and quantity of sample that should be weighed. Then the buffer solution can be prepared easily by adding water to the specified quantity of sample.

Preparation example: When weighing and preparing 50 mM of di-sodium hydrogen phosphate, 2-hydrate and 50 mM of sodium dihydrogenphosphate, 2-hydrate in order to prepare 3 L of 100 mM phosphoric acid (sodium) buffer solution at pH = 2.1:

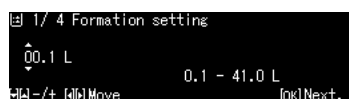
Example of preparation by AP series



Select the buffer solution mode.



Specify the type and quantity.



Displays the name and quantity of sample.



Prepare as instructed on screen.

Complete buffer solution



Number		Buffer solution preparation list	
1	100 mM	phosphoric acid (sodium)	pH = 2.1
2	10 mM	phosphoric acid (sodium)	pH = 2.6
3	50 mM	phosphoric acid (sodium)	pH = 2.8
4	100 mM	phosphoric acid (sodium)	pH = 6.8
5	10 mM	phosphoric acid (sodium)	pH = 6.9
6	20 mM	citric acid (sodium)	pH = 3.1
7	20 mM	citric acid (sodium)	pH = 4.6
8	10 mM	tartaric acid (sodium)	pH = 2.9
9	10 mM	tartaric acid (sodium)	pH = 4.2
10	20 mM	acetic acid (ethanolamine)	pH = 9.6
11	100 mM	acetic acid (sodium)	pH = 4.7
12	100 mM	boric acid (potassium)	pH = 9.1
13	100 mM	boric acid (sodium)	pH = 9.1

* Results can be printed with date/time and user ID.

Sample Preparation **NEW**

(AP-W Series only)

When preparing a standard solution from a particular component, the standard powder for this component will be a hydrochloride or a hydrate. Preparing a standard solution of the target component at a desired concentration requires difficult calculations prior to weighing it. With the AP series, however, the required weight value is automatically calculated, so it can be weighed without performing manual calculations.

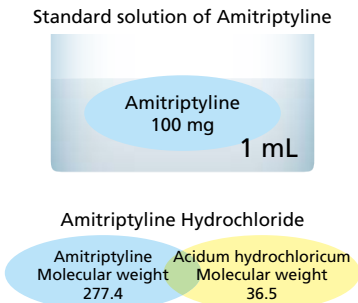
Example of preparation by AP series

Weigh 25 mg Amitriptyline to make a standard solution

Standard sample of Amitriptyline is Amitriptyline Hydrochloride.
Calculation is essential to determine part of Acidum hydrochloricum by molecular weight in order to make a 100 mg/mL Amitriptyline solution.
Molecular weight of Amitriptyline: 277.4
Molecular weight of Acidum hydrochloricum: 36.5
Molecular weight of Amitriptyline Hydrochloride: $277.4 + 36.5 = 313.9$

To compare the molecular weight of Amitriptyline Hydrochloride with Amitriptyline, the following calculation is necessary.
 $313.9 / 277.4 = 1.132$

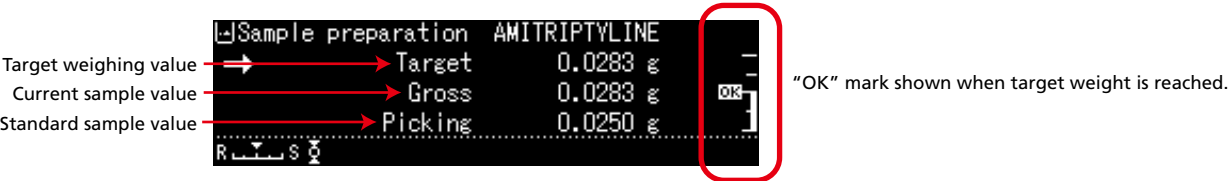
The molecular weight of Amitriptyline Hydrochloride is 1.132 times of Amitriptyline.
So, if 25mg of Amitriptyline is used, it follows that the weight of Amitriptyline Hydrochloride should be:
 $25 \text{ mg} \times 1.132 = 28.3 \text{ mg}$.
Hence, 28.3 mg of Amitriptyline Hydrochloride is needed to make the correct standard solution.



No need for manual calculation

AP series can automatically calculate the sampling weight using the molecular amount of the standard sample, molecular weight of unnecessary sample, and the target value in order to reach the correct concentration solution.

Just weigh the target weight value on display and the target weight of the standard sample can be obtained.



Checking weighing conditions on the same display

Save Your Operation

USB Offers Greater Expandability **NEW**

(USB host: AP-W Series only)

Equipped with an RS-232C connector, a USB device and a USB host as standard. You can now simultaneously send output to both a computer and printer or connect a USB flash drive, a barcode reader, or an external numeric keypad.

Transcription errors can be avoided and data can be recorded without a computer.



USB and RS-232C are standard

USB host is available for AP-W Series

USB flash drive

Connecting a USB memory device allows you to record large amounts of weighing data in CSV format. Used in combination with the interval output function, it enables recording of long-term changes over time.

Example of a record:

File name
Date and time
Weighing value

* The information saved will differ depending on the function used.



Display capture function

Weighing display can be recorded into USB memory in BMP format. User name, date/time, and setting can be shown with display information.

The user name, time, measurement conditions, pass/fail judgments, and other information displayed on screen can be saved as is, enabling the recording of measurements and checks after measurements.



Numeric keypad

Connecting a common external numeric keypad makes it easier to enter numeric values. This is especially useful for entering the mass value of weights, setting upper/lower limit values for the comparator function, or entering the sample count during piece counting mode.

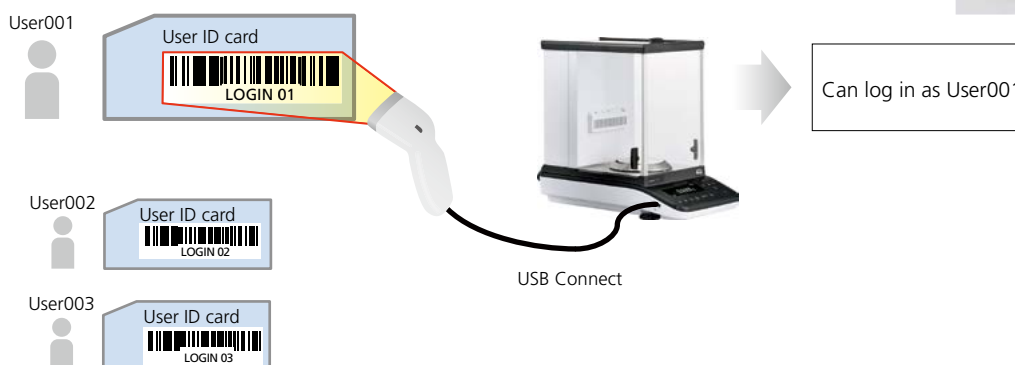


Barcode reader

A barcode reader can be connected. Simply reading a barcode makes it possible to input user ID/Password. It is possible to manage sample IDs using barcodes.



An example of login by barcode



An ID and password are needed to log in to the AP series if protected access is activated. With the barcode, an operator can log in by scanning the barcode instead of inputting an ID and password.

* The latest information can be seen from the Shimadzu website (<https://www.shimadzu.com/an/balance/>)

Equipped with USB as standard.*1
Includes many diverse functions to support users.

*1 AP-W Series only

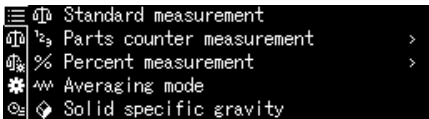
Easy-to-Read Organic EL Display (All models)

Because the pixel elements in the organic electroluminescence display emit light, the screen can be seen clearly even in dark locations. Multi-language display capability*2 provides a more intuitive operating interface. A wider viewing angle has also improved the visibility of measurement values, which helps increase the efficiency of measuring operations.

*2 English, Chinese and Japanese



Clearly visible from the side



English example

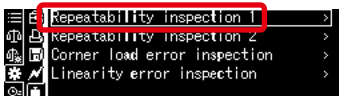
Exceptional Visibility

The visibility remains the same even when viewed from different angles. The viewing angle is a wide expanse of ± 85 degrees, both vertically and horizontally. That means the display is clearly visible even when working beside the balance. A high-resolution dot-matrix display makes it easy to read detailed text.

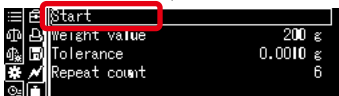
Periodic Inspection Support Function NEW (AP-W/AP-X Series only)

AP series supports periodic inspections. The function allows inspection of repeatability, corner load error, and linearity by simply following instructions displayed on the screen.

Example of printing



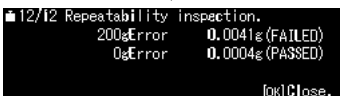
Select the inspection mode.



In this case, repeatability inspection is selected.



Place standard weights as instructed.



Results are displayed.



Printing sample

REPEATABILITY	
LOAD	= 150 g
MPE	= 0.0010 g
N001	
IL	= 150.0000 g
I0	= 0.0000 g
N002	
IL	= 149.9999 g
I0	= -0.0001 g
N003	
IL	= 149.9999 g
I0	= -0.0001 g
N004	
IL	= 149.9999 g
I0	= 0.0000 g
N005	
IL	= 149.9999 g
I0	= 0.0000 g
N006	
IL	= 149.9999 g
I0	= 0.0000 g
TEST RESULTS	
LOAD	= 0.0001 g (PASSED)
ZERO	= 0.0001 g (PASSED)

IL: Loaded weight
I0: Zero value

And more...

Wide Variety of Functions to Support Users

Smart Settings

(All models)

Response and stability settings can be changed during measurements with a single touch. Changing the settings for different applications can make it even easier to use.



User-friendly arrow keys

The indicator is operated using the left and right arrow keys. Moving the setting toward [R] prioritizes response, which makes it easier to operate the balance. Conversely, moving it toward [S] makes it easier to stabilize weight values, which can improve readability in environments with vibration.



Patented

Moving it left prioritizes response and moving it right prioritizes stability. Five setting levels are available.

Specific Gravity Measurement

(All models)

In combination with an optional specific gravity measurement kit, the balance can be used to measure specific gravity. Operations are simplified by a text-based navigation function. By using sinkers, the specific gravity of liquid can be measured as well. This allows measuring the specific gravity of metals, rubbers, plastics, and other materials easily.



First measure the empty weight.



Then place it in the container filled with water, as instructed on the screen.



The specific gravity value is displayed using simple steps.



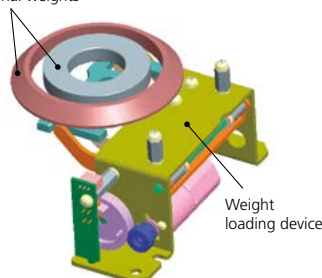
For Better Weighing Results

(AP-W/AP-X Series only)

- Two internal weights provided

0.01 mg models are calibrated at 2 points with the internal weights (weight value and 1/2 value).

Two internal weights

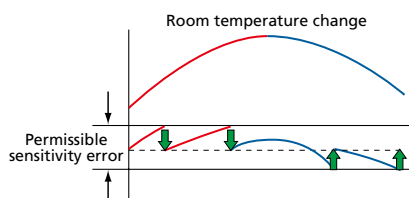


Weight loading device

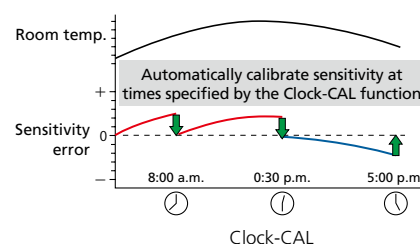
- Includes Perfect Self Calibration (PSC) function

The analytical balance automatically detects any temperature changes that could affect sensitivity and automatically starts calibration.

The Clock-CAL function enables automatic calibration at a pre-specified time (for example, before starting work, during lunch, or after work hours).



PSC (Perfect Self Calibration)



Clock-CAL

The Reason the AP Holder in Combination with the Ionizer Can Eliminate the Influence of Static Electricity on the Weight Value

Why is the AP Holder needed as a countermeasure to static electricity?

If the AP Holder and the STABLO-AP ionizer are used together, static electricity can be quickly removed from the entire test chamber, including the surfaces of glass containers, which helps to decrease the weighing time and improve reliability.

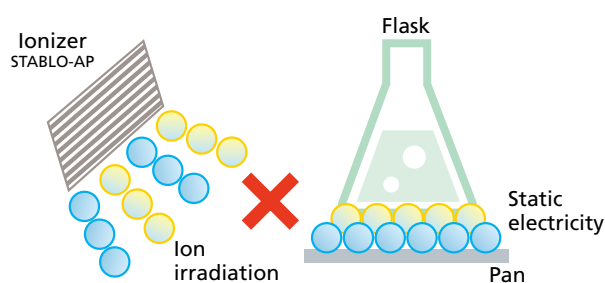


AP Holder

Example of Removing Static Electricity from a Flask

NG

The conical flask is directly placed on the pan.

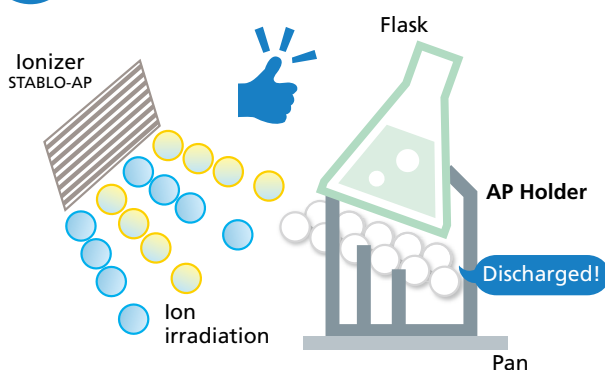


Ions emitted from the ionizer cannot reach the bottom of the flask, so removal of static charge from the bottom of the flask is insufficient. Therefore, Coulomb forces act between the surrounding metal parts and the windshield door, which affects the weight value.

The bottom of the flask is in close contact with the pan, so removal of the static charge is obstructed, leading to an unstable weight value.

OK

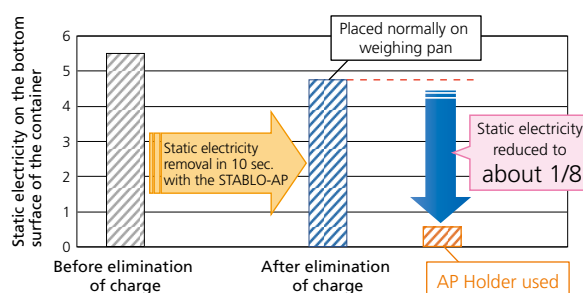
The conical flask is tilted using the AP Holder and placed on the pan.



Using the AP Holder to separate the flask from the pan, the ions supplied by the ionizer reach the locations where there is static charge on the bottom of the flask. This improves the neutralization effect and results in a stable weight value.

The AP Holder can hold the container in a tilted position, so the charge can be reliably removed from the bottom of the container being mounted on the AP Holder.

Effect of AP Holder on Eliminating Static Charge
(measurement is carried out by Shimadzu)



Static electricity on the bottom surface of the charged measuring flask (capacity 100 mL) measured with a surface potentiometer

AP Series Specifications

W Series Analytical Balances

Series	W Series						
Model	AP225W	AP135W	AP125WD	AP225WD	AP124W	AP224W	AP324W
Capacity	220 g	135 g	120 g / 52 g	220 g / 102 g	120 g	220 g	320 g
Minimum Display	0.01 mg		0.1 mg / 0.01 mg		0.1 mg		
Calibration Weight	Built-in*1						
External Calibration Weight Range for Span Calibration	95 to 220.00090 g (200 g)	45 to 135.00090 g (100 g)	45 to 120.00090 g (100 g)	95 to 220.00090 g (200 g)	45 to 120.009 g (100 g)	95 to 220.009 g (200 g)	95 to 320.009 g (300 g)
Repeatability (Standard deviation)	0.015 mg (to 20 g) 0.03 mg (to 100 g) 0.05 mg (to weighing capacity)	0.05 mg	0.1 mg / 0.02 mg	0.1 mg / 0.05 mg	0.1 mg		0.15 mg
Repeatability (for Low Loads)*2	0.01 mg (5 g low loads)	0.015 mg (5 g low loads)			0.1 mg (5 g low loads)	0.1 mg (10 g low loads)	0.1 mg (20 g low loads)
Minimum Weight*2	20 mg				200 mg		
Linearity	±0.1 mg		±0.2 mg / ±0.05 mg	±0.2 mg / ±0.1 mg	±0.2 mg		±0.3 mg
Response Time for Trace Measurements*3				2 sec.			
Response Time*4	8 sec.		2 sec. / 8 sec.		2 sec.		
USB Host (Type A)				Included			
USB Device (Type B)				Included			
Recipe Compounding				Included			
HPLC Buffer Solution Preparation				Included			
mol Conversion Function				Included			
Sample (Concentration) Preparation				Included			
Inspection Support Function				Included			
Clock-CAL				Included			
Ionizer				Optional			
Operating Temperature/Humidity Range				5 to 40°C 20 to 85%*5			
Sensitivity Stability Against Temperature Range				±2 ppm/°C (10 to 30°C)			
Pan Size				ø91 mm			
Body Dimensions	Approx. W212 × D411 × H345 mm (power supply unit included)			Approx. W212 × D367 × H345 mm			
Weight	Approx. 7.9 kg			Approx. 7.0 kg			
Display				OEL display (dot matrix)			
Input/Output Terminal	RS-232C (D-sub 9P plug)			USB host (Type A)	USB device (Type B)	Ionizer	

X Series / Y Series Analytical Balances

Series	X Series			Y Series		
Model	AP124X	AP224X	AP324X	AP124Y	AP224Y	AP324Y
Capacity	120 g	220 g	320 g	120 g	220 g	320 g
Minimum Display	0.1 mg					
Calibration Weight	Built-in			No		
External Calibration Weight Range for Span Calibration	45 to 120.009 g (100 g)	95 to 220.009 g (200 g)	95 to 320.009 g (300 g)	45 to 120.009 g (100 g)	95 to 220.009 g (200 g)	95 to 320.009 g (300 g)
Repeatability (Standard deviation)	0.1 mg		0.15 mg	0.1 mg		0.15 mg
Repeatability (for Low Loads)* ²	0.1 mg (5 g low loads)	0.1 mg (10 g low loads)	0.1 mg (20 g low loads)	0.1 mg (5 g low loads)	0.1 mg (10 g low loads)	0.1 mg (20 g low loads)
Minimum Weight* ²	200 mg					
Linearity	±0.2 mg		±0.3 mg	±0.2 mg		±0.3 mg
Response Time for Trace Measurements* ³	2 sec.					
Response Time* ⁴	2 sec.					
USB Host (Type A)	Not Included					
USB Device (Type B)	Included					
Recipe Compounding	Not Included					
HPLC Buffer Solution Preparation	Not Included					
mol Conversion Function	Included			Not Included		
Sample (Concentration) Preparation	Not Included					
Inspection Support Function	Included			Not Included		
Clock-CAL	Included			Not Included		
Ionizer	Optional			Not Included		
Operating Temperature/Humidity Range	5 to 40°C 20 to 85%* ⁵					
Sensitivity Stability Against Temperature Range	±2 ppm/°C (10 to 30°C)					
Pan Size	ø91 mm					
Body Dimensions	Approx. W212 × D367 × H345 mm					
Weight	Approx. 7.0 kg			Approx. 6.5 kg		
Display	OEL display (dot matrix)					
Input/Output Terminal	RS-232C (D-sub 9P plug)	USB device (Type B)	Ionizer	RS-232C (D-sub 9P plug)	USB device (Type B)	

*1 Minimum display 0.01 mg models include two internal weights (see page 22 for details).

*2 Be compliant with USP Chapter 41. This is the tested value by the weight of the balance's capacity 5% (or 5 grams' weight). In the case of the AP225W, the results are for tests carried out with the optional internal windbreak plate applied. The minimum weight value is affected by the installation environment, so it is necessary to measure it in the actual environment of use.

*3 The response time for displaying 90% of the added sample amount value in trace measurements (from 1 mg).

*4 The response time value is typical.

*5 Non-condensing.

AP Series

0.01 mg model

AP225W
AP135W

0.01 mg / 0.1 mg model

AP225WD
AP125WD



* Power supply unit included.

0.1 mg model

AP124W AP124X AP124Y
AP224W AP224X AP224Y
AP324W AP324X AP324Y



STABLO-AP Ionizer



EP-100 Electronic Printer



EP-110 Electronic Printer
(Multifunction printer with liquid crystal display)



SMK-601
Specific Gravity Measurement Kit



Internal Windbreak Plate



AP Holder

Optional Accessories

Static Electricity Remover STABLO-AP Ionizer
Electronic Printer EP-100
Electronic Printer EP-110
Label Roll Paper for EP-100/110 (10 Rolls)
Specific Gravity Measurement Kit SMK-601
Display Protective Cover (Set of 5)

USB Cable
RS-232C Cable
AC Adapter (Standard accessory)
Internal Windbreak Plate
RS-I/O Interface Cable
AP Holder (Standard accessory for AP225W)

Electronic Balances

UP^{series}

Multi Functional Top-Loading Balances

Top-loading Balances with the Fastest Response Performance in Its Class



Large Pan with 0.01 g
Minimum Display Value
UP2202X UP2202Y
UP4202X UP4202Y
UP6202X UP6202Y



Large Pan with 0.1 g
Minimum Display Value
UP4201X UP4201Y
UP8201X UP8201Y



Small Pan with 0.001 g
Minimum Display Value
UP223X UP223Y UP423X UP423Y
UP623X UP623Y UP823X UP823Y
UP1023X UP1023Y

* Windbreak can be removed.



Small Pan with 0.01 g
Minimum Display Value
UP422X UP422Y
UP822X UP822Y



PSC (Perfect Self-Calibration) (UP-X series only)

Automatically calibrates the balance if an ambient temperature change occurs that could affect sensitivity.



Clock-CAL (UP-X series only)

This automatically calibrates the balance at pre-specified times (such as before starting work, during lunch, or after work hours).



Internal Calibration (UP-X series only)

Enables calibration by pressing a single key with one finger. That ensures calibration can be performed whenever needed.



Printout with Date and Time

If a printer (optional) is connected, then data can be marked with the date and time. Calibration results can also be marked with date and time, which is perfect for managing measurements or establishing traceability as required by GLP/GMP/ISO 9001 standards.



ISO Calibration Report

Simply connect an optional printer to automatically print out which balance was calibrated when, and the calibration results. Absolutely no troublesome settings are required. Furthermore, the current date and time can be printed at anytime during measurement.



Backlight LCD

This backlit LCD can be clearly read even in the darkest of environments.



Checkweighing

By specifying upper and lower limit values in advance, the comparator indicates whether the sample weight passes the criteria, is too heavy, or is too light.



Comparator Output

Acceptable, high, or low weight results or other pass/fail results can be indicated by an audible sound or output externally as a contact signal. (Optional comparator audible indicator or relay output interface required.)



Measuring Carats

The balance supports measurements in carats, *momme*, percent, and a variety of other measurement units. (The unit *momme* is only used for measuring pearls.)



Computer Connection Function

Systems can be connected to a computer via an RS-232C cable or using a USB–serial adapter kit. For more details, visit the Shimadzu website.



Piece Counting

A built-in piece counting function enables the balance to be used as a parts counter (counting balance).



Specific Gravity Measurement Function

The built-in specific gravity measurement function is based on the liquid immersion method. By installing the specific gravity measurement kit (optional), the balance can be used as a hydrometer.



UP-X Series



UP-Y Series



UP-X Series

Model name	UP223X	UP423X	UP623X	UP823X	UP1023X	UP2202X	UP4202X	UP6202X	UP422X	UP822X	UP4201X	UP8201X
Capacity	220 g	420 g	620 g	820 g	1020 g	2200 g	4200 g	6200 g	420 g	820 g	4200 g	8200 g
Minimum Display	0.001 g	0.001 g	0.001 g	0.001 g	0.001 g	0.01 g	0.01 g	0.01 g	0.01 g	0.01 g	0.1 g	0.1 g
Pan Size (mm)		Approx. 108 × 105				Approx. 170 × 180			Approx. 108 × 105		Approx. 170 × 180	

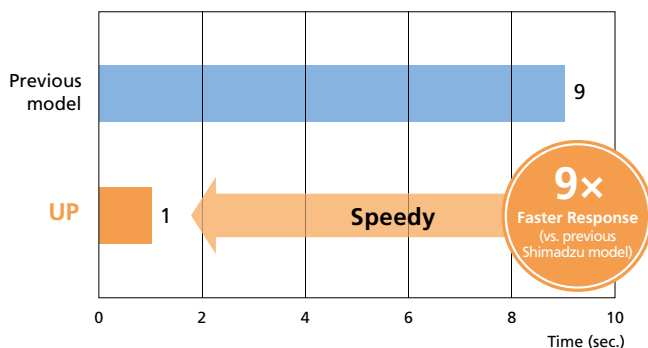
UP-Y Series

Model name	UP223Y	UP423Y	UP623Y	UP823Y	UP1023Y	UP2202Y	UP4202Y	UP6202Y	UP422Y	UP822Y	UP4201Y	UP8201Y
Capacity	220 g	420 g	620 g	820 g	1020 g	2200 g	4200 g	6200 g	420 g	820 g	4200 g	8200 g
Minimum Display	0.001 g	0.001 g	0.001 g	0.001 g	0.001 g	0.01 g	0.01 g	0.01 g	0.01 g	0.01 g	0.1 g	0.1 g
Pan Size (mm)		Approx. 108 × 105				Approx. 170 × 180			Approx. 108 × 105		Approx. 170 × 180	

Response for Weighing Minute Quantities

Large-pan model with 0.01 g minimum display value

Small-pan model with 0.001 g minimum display value



Measurement Conditions:

Large-pan model with 0.01 g minimum display value and 0.1 g load

Small-pan model with 0.001 g minimum display value and 0.01 g load

Optional Accessories

Description
Static Electricity Remover STABLO-AP Ionizer
Electronic Printer EP-100
Electronic Printer EP-110
RS-232C Cable 25P-9P (1.5 m)
USB-Serial Replacement Kit
Application Keyboard AKB-301
Windbreak Set *1
Windbreak Set (For large pan) *2
Glass Windbreak WBC-102
Large Size Windbreak WBC-502
In-use Protective Cover (3 pcs) (For large-pan models with 0.01 g min. display value)
In-use Protective Cover (5 pcs) (For large-pan models with 0.1 g min. display value)
In-use Protective Cover (5 pcs) (For small-pan model)
In-use Protective Cover (5 pcs) (For display and key part)
Animal Bucket Set (For large-pan models with 0.01 g minimum display value)
Animal Bucket Set (For large-pan models with 0.1 g minimum display value)
Specific Gravity Measurement Kit SMK-101A (For large-pan model) (Up to approximately 2 kg for weighing capacity)
Specific Gravity Measurement Kit SMK-102 (For small-pan model)
Remote Display RDB-201
Remote Display RDB-202
Relay Output Interface IFB-RY1
RY1 Connection Cable
AC Adapter *3

*1 Included standard with small pan models with 0.001 g minimum display value.

*2 Included standard with large pan models with 0.01 g minimum display value.

*3 Included standard with main unit.